Part 1. Exploratory Data Analysis

Group 3

Dataset scope and coverage

We compared 119,390 bookings at a Lisbon city hotel and an Algarve resort to see how demand changed through the year. Arrivals ran from July 2015 to August 2017, making 2016 the only complete year. We therefore compared the hotels month by month, since annual totals would give 2016 an advantage simply because it covers more months. We grouped the 32 fields (15 categorical and 17 numerical) by what they tell us about the guests, their stay and the booking itself (Figures 1–4).

Data cleaning criteria

When we added up the adults, children and babies on each booking, we found 180 with nobody listed, so we excluded them. We also removed a negative nightly rate (−6.38), a rate of 5,400 above our 5,000 cutoff, and two bookings listing nine or more babies, which we judged to be recording errors. Five more had a missing child count or an undefined market segment or booking channel. Removing these 189 rows left 119,201 bookings (Figure 5).

A missing entry meant different things depending on the field. We recorded “No agent” or “No company” when no intermediary was listed and treated 1,169 undefined meal entries as “No meal package”. For 478 bookings we had no way to establish the guest’s country, so we used “Unknown”. We interpreted CN as Canada in 1,279 bookings because China already appeared as CHN and Canada was otherwise absent, although the records cannot confirm that assumption.

We kept unusual entries that could represent real bookings, including zero prices, zero nights, children booked without adults, large groups and 27 check-out date discrepancies. Of the original rows, 31,994 repeated an earlier row exactly. Without a booking identifier, we could have erased separate reservations by removing every match, so we kept the 31,980 repeats that remained after cleaning.

Booking patterns and analytical decisions

Looking at how far ahead people booked, we saw that a few very early reservations pulled the mean up to 104 days. We chose the median of 69 days because it better describes a typical booking (Figure 6). Most stays lasted one to four nights, with peaks at one and two weeks, and Portugal accounted for 40.7% of bookings (Figures 7–8). Prices varied most at the resort, where the median nightly rate rose from 46.8 in November to 180.4 in August (Figures 9–10).

In the raw data, 37.0% of bookings were cancelled, but the rate ranged from 9.6% for bookings made within a week of arrival to 67.7% for those made more than a year ahead. For non-refundable bookings it reached 99.4%, a figure repeated entries may have inflated (Figure 11). We therefore kept booking lead time alongside cancellations, and calculated the arrival date, total guests and total nights for each booking so we could compare demand by hotel, country and price. We left final reservation status out of any prediction because it already tells us whether the booking was cancelled. These findings describe a historical period. For current planning, we would need fresh bookings daily or monthly, depending on the decision.

Figures

Field	Meaning	Data type	Order	Temporal or geographic
hotel	Hotel of the booking: Resort Hotel (Algarve) or City Hotel (Lisbon). Added when the two datasets of the paper were joined	Categorical, nominal	No order	Geographic (hotel location)
is_canceled	The booking was cancelled (1) or not (0)	Categorical, nominal (binary)	No order	Neither
lead_time	Days between the booking entering the hotel system and the arrival date	Numerical, discrete, ratio	Sequential	Temporal (duration)
arrival_date_year	Year of the arrival date	Numerical, discrete, interval	Sequential	Temporal
arrival_date_month	Month of the arrival date (written as the month name)	Categorical, ordinal	Cyclic	Temporal
arrival_date_week_number	Week number of the arrival date in its year	Numerical, discrete, interval	Cyclic	Temporal
arrival_date_day_of_month	Day of the month of the arrival date	Numerical, discrete, interval	Cyclic	Temporal
stays_in_weekend_nights	Weekend nights (Saturday or Sunday) stayed or booked to stay	Numerical, discrete, ratio	Sequential	Temporal (duration)
stays_in_week_nights	Week nights (Monday to Friday) stayed or booked to stay	Numerical, discrete, ratio	Sequential	Temporal (duration)
adults	Number of adults	Numerical, discrete, ratio	Sequential	Neither
children	Number of children (paying and non-paying)	Numerical, discrete, ratio	Sequential	Neither
babies	Number of babies	Numerical, discrete, ratio	Sequential	Neither
meal	Meal package: SC or Undefined = no meal, BB = bed and breakfast, HB = half board, FB = full board	Categorical, ordinal (meals included)	Sequential	Neither
country	Country of origin of the guest, recorded as a three-letter abbreviation	Categorical, nominal	No order	Geographic (guest origin)
market_segment	Market segment of the booking (TA = travel agents, TO = tour operators)	Categorical, nominal	No order	Neither
distribution_channel	Channel used to make the booking (TA = travel agents, TO = tour operators, GDS = global distribution system)	Categorical, nominal	No order	Neither

Figure 1. What fields 1–16 measure and how we classified them.

Field	Meaning	Data type	Order	Temporal or geographic
is_repeated_guest	The booking name belongs to a repeated guest (1) or not (0)	Categorical, nominal (binary)	No order	Neither
previous_cancellations	Earlier bookings of the same customer that were cancelled	Numerical, discrete, ratio	Sequential	Neither
previous_bookings_not_canceled	Earlier bookings of the same customer that were not cancelled	Numerical, discrete, ratio	Sequential	Neither
reserved_room_type	Code of the room type reserved (letters, anonymised)	Categorical, nominal	No order	Neither
assigned_room_type	Code of the room type assigned; can differ from the reserved one (overbooking, guest request)	Categorical, nominal	No order	Neither
booking_changes	Changes made to the booking until check-in or cancellation	Numerical, discrete, ratio	Sequential	Neither
deposit_type	Deposit to guarantee the booking: No Deposit, Non Refund (covers the whole stay) or Refundable (covers part of it)	Categorical, nominal	No order	Neither
agent	ID of the travel agency that made the booking; NULL = not made by an agency	Categorical, nominal (identifier)	No order	Neither
company	ID of the company that made or pays the booking; NULL = no company	Categorical, nominal (identifier)	No order	Neither
days_in_waiting_list	Days the booking waited before it was confirmed to the customer	Numerical, discrete, ratio	Sequential	Temporal (duration)
customer_type	Type of booking: Transient, Transient-Party, Contract or Group	Categorical, nominal	No order	Neither
adr	Average daily rate: lodging revenue of the booking divided by its number of nights	Numerical, continuous, ratio	Sequential	Neither
required_car_parking_spaces	Car parking spaces the customer asked for	Numerical, discrete, ratio	Sequential	Neither
total_of_special_requests	Special requests made by the customer (for example a twin bed or a high floor)	Numerical, discrete, ratio	Sequential	Neither
reservation_status	Last status of the booking: Canceled, Check-Out or No-Show. Repeats is_canceled	Categorical, nominal	No order	Neither
reservation_status_date	Date on which the last status was set	Numerical, discrete, interval (date)	Sequential	Temporal

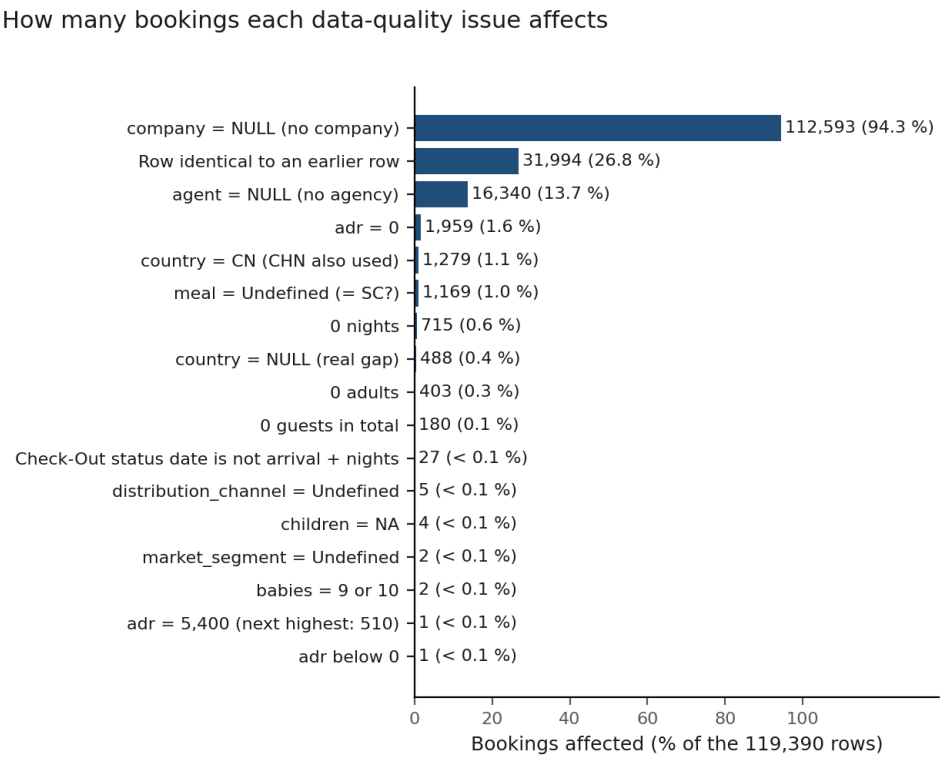
Figure 2. What fields 17–32 measure and how we classified them.

Field	Range or domain (raw data)	Units	Precision needed	# Missing	Missing placeholder
hotel	2: City Hotel, Resort Hotel	No unit	Label	0	—
is_canceled	0 = no, 1 = yes	No unit	Count; rate in % with 1 decimal	0	—
lead_time	0 to 737	Days	Whole days	0	—
arrival_date_year	2015 to 2017	Year	Year	0	—
arrival_date_month	January to December (12 months)	Month	Month	0	—
arrival_date_week_number	1 to 53	Week of the year	Week	0	—
arrival_date_day_of_month	1 to 31	Day of the month	Day	0	—
stays_in_weekend_nights	0 to 19	Nights	Whole nights	0	—
stays_in_week_nights	0 to 50	Nights	Whole nights	0	—
adults	0 to 55	Persons	Whole persons	0	—
children	0 to 10	Persons	Whole persons	4	NA
babies	0 to 10	Persons	Whole persons	0	—
meal	5: BB, HB, SC, Undefined, FB	No unit	Label	1,169	Undefined
country	177 codes (most frequent: PRT, GBR, FRA)	No unit	Country	488	NULL
market_segment	8: Online TA, Offline TA/TO, Groups, Direct, Corporate, Complementary, Aviation, Undefined	No unit	Label	2	Undefined
distribution_channel	5: TA/TO, Direct, Corporate, GDS, Undefined	No unit	Label	5	Undefined

Figure 3. Values, units, precision and missing entries for fields 1–16.

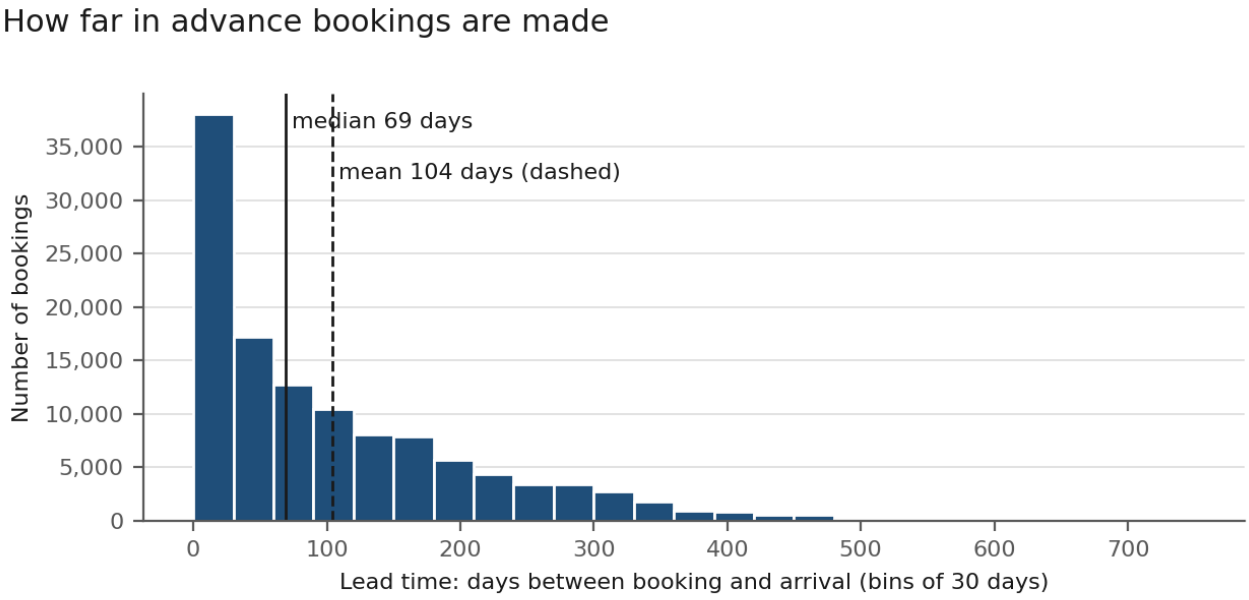
Field	Range or domain (raw data)	Units	Precision needed	# Missing	Missing placeholder
is_repeated_guest	0 = no, 1 = yes	No unit	Count; share in % with 1 decimal	0	—
previous_cancellations	0 to 26	Bookings	Whole bookings	0	—
previous_bookings_not_canceled	0 to 72	Bookings	Whole bookings	0	—
reserved_room_type	10: A, D, E, F, G, B, C, H, P, L	No unit	Label	0	—
assigned_room_type	12: A, D, E, F, G, C, B, H, I, K, P, L	No unit	Label	0	—
booking_changes	0 to 21	Changes	Whole changes	0	—
deposit_type	3: No Deposit, Non Refund, Refundable	No unit	Label	0	—
agent	333 IDs, from 1 to 535	No unit	Label	16,340	NULL
company	352 IDs, from 6 to 543	No unit	Label	112,593	NULL
days_in_waiting_list	0 to 391	Days	Whole days	0	—
customer_type	4: Transient, Transient-Party, Contract, Group	No unit	Label	0	—
adr	-6.38 to 5400	Money per night (currency not stated; probably euros)	Whole euros (file has 2 decimals)	0	—
required_car_parking_spaces	0 to 8	Parking spaces	Whole spaces	0	—
total_of_special_requests	0 to 5	Requests	Whole requests	0	—
reservation_status	3: Check-Out, Canceled, No-Show	No unit	Label	0	—
reservation_status_date	2014-10-17 to 2017-09-14	Date (YYYY-MM-DD)	Day	0	—

Figure 4. Values, units, precision and missing entries for fields 17–32.



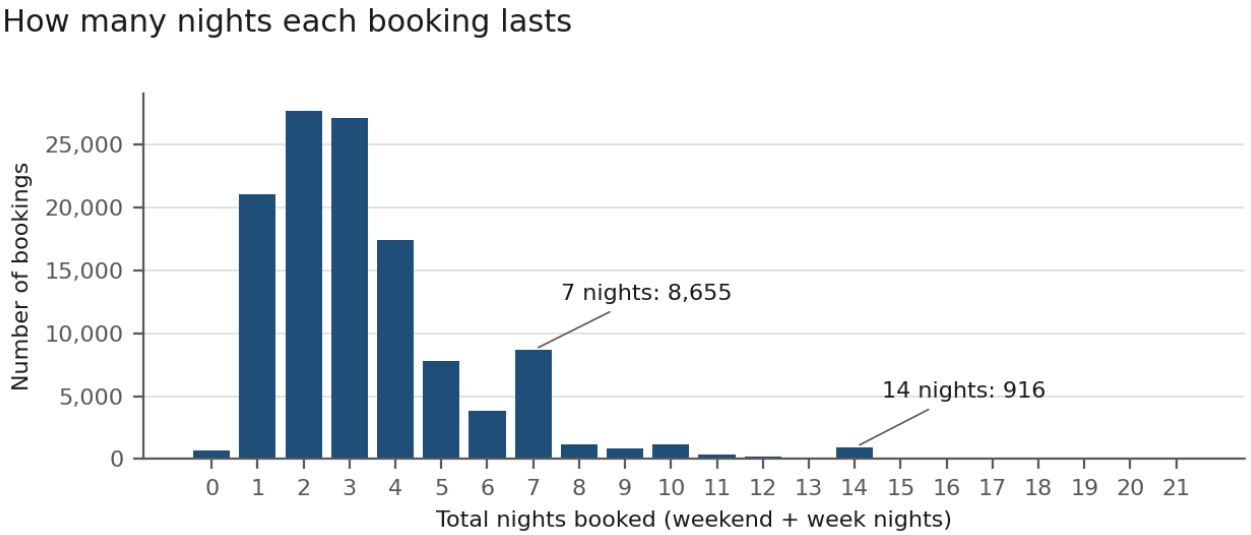
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file.

Figure 5. Bookings affected by each data quality issue.



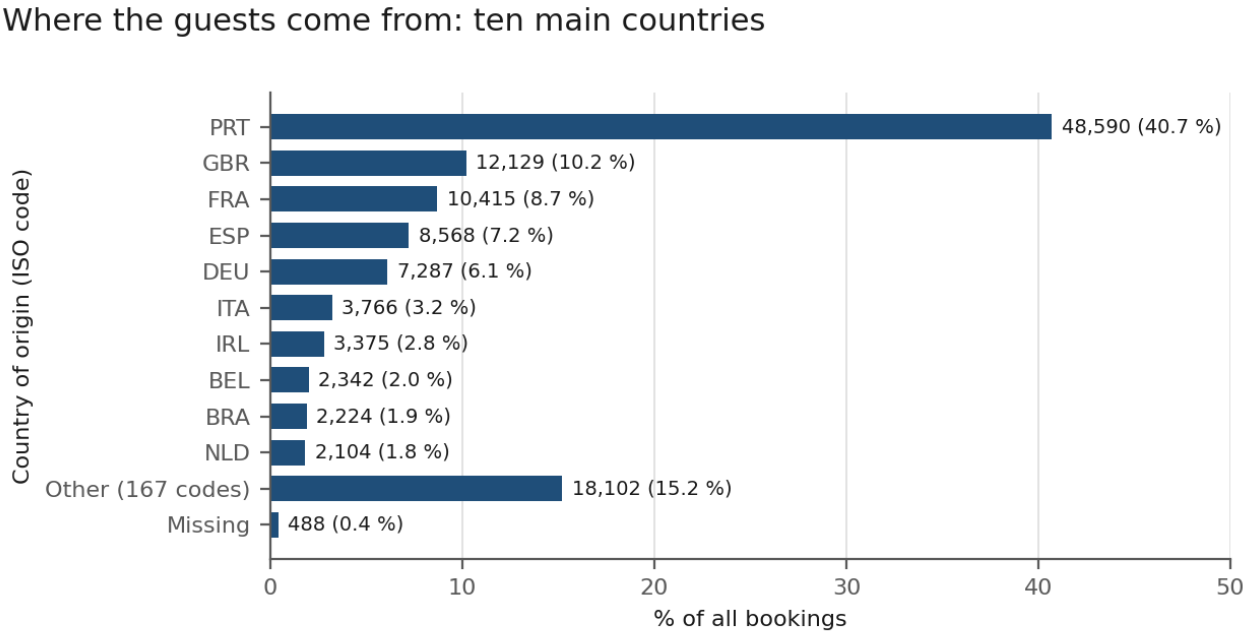
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 6. Days between booking and arrival. Early bookings pull the mean above the median.



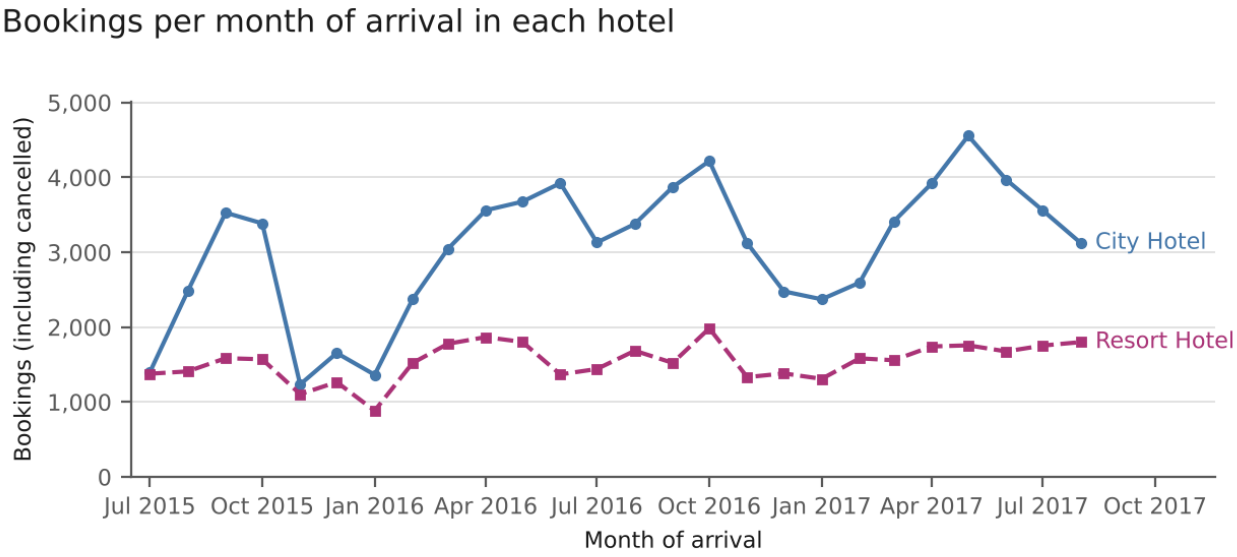
Not shown: 162 bookings of more than 21 nights (maximum 69 nights).
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 7. Nights per booking. Stays cluster at seven and fourteen nights.



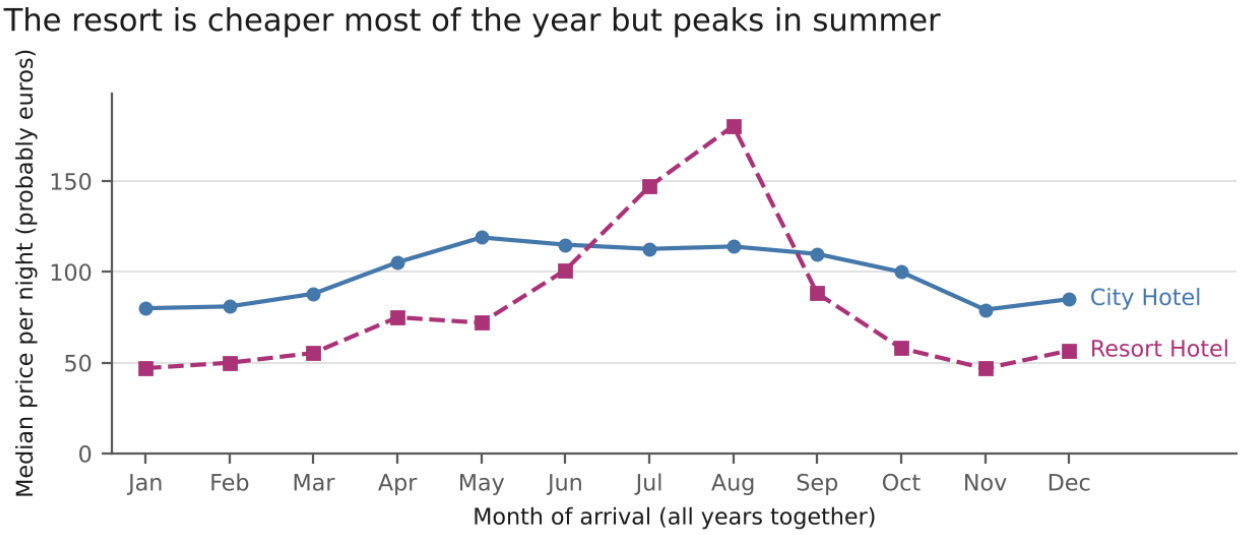
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 8. The ten most frequent countries of origin in the booking records.



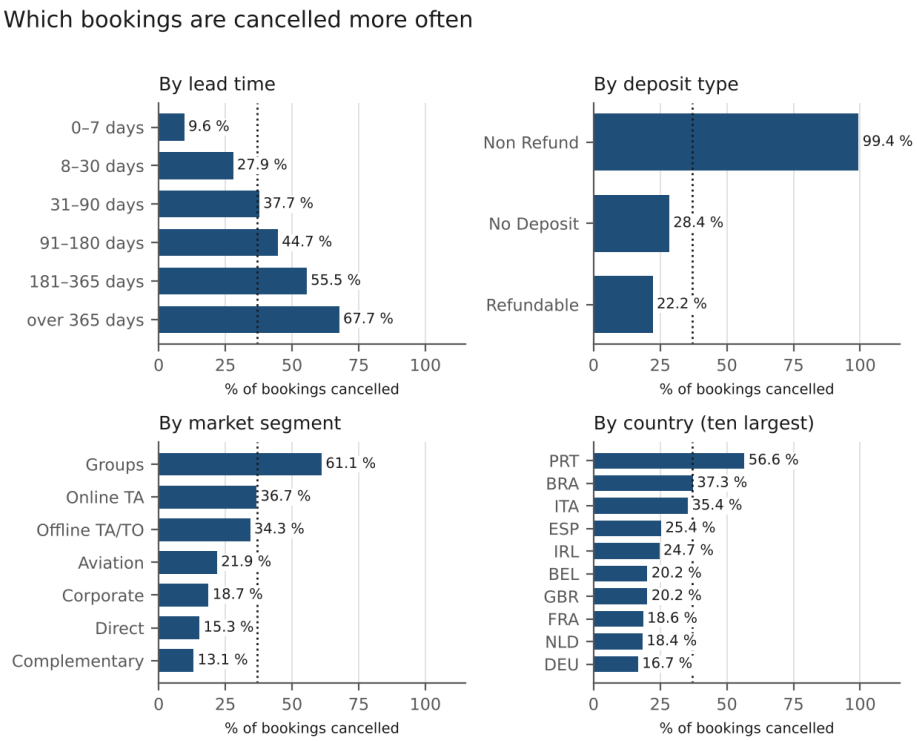
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 9. Monthly bookings at the city hotel and the resort.



Only stays that were not cancelled and have a price above 0.
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 10. Median nightly rate by month at each hotel.



Dotted line: rate over all bookings (37.0 %). Raw data, duplicates included.
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 11. Cancellation rates by booking lead time, deposit, market segment and country.

Data source

Antonio, N., de Almeida, A. and Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*, 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126